

ECO 310 - Solution Set 2

Due: Friday, Feb 27

1. Income and Substitution Effects

Assume perfect substitute preferences, with $u(x_1, x_2) = x_1 + x_2$.

- (a) Calculate the compensated demand bundle $(x_1^c(p_1, p_2, u^*), x_2^c(p_1, p_2, u^*))$. (Hint: you should have a piecewise function, depending on which good is cheaper).

Answer: if $p_1 < p_2$, the consumer optimally buys good 1 only; then plugging $x_2 = 0$ into the utility constraint $x_1 + x_2 = u^*$ yields $x_1 = u^*$. By the same logic, if good 2 is cheaper, the consumer buys good 2 only, and needs u^* units of it to achieve his utility goal. If $p_1 = p_2$, any combination with $x_1 = x_2 = u^*$ achieves utility u^* in the cheapest way. So,

$$(x_1^c, x_2^c) = \begin{cases} (u^*, 0) & \text{if } p_1 < p_2 \\ (0, u^*) & \text{if } p_2 < p_1 \end{cases}$$

(And any $(\Delta, u^* - \Delta)$ for $\Delta \in [0, u^*]$ when $p_1 = p_2$; no points off if you didn't do this).

- (b) Using your answer to (a), calculate the expenditure function, $e(p_1, p_2, u^*)$

Answer: this is the cost of the bundle in (a), ie the one that most cheaply attains utility u^* . So

$$e(p_1, p_2, u^*) = \begin{cases} p_1 u^* & \text{if } p_1 < p_2 \\ p_2 u^* & \text{if } p_2 < p_1 \end{cases}$$

And either of these if $p_1 = p_2$. More simply, the expenditure function is $\min\{p_1, p_2\} \cdot u^*$

- (c) Suppose that initially, $p_1 = 1$ and $p_2 = 2$ and $m = 10$, so that this consumer buys the bundle $(10, 0)$ and earns utility 10.

- i. Using your answer to (a), calculate the substitution effect if p_1 increases to 1.5.

Answer: Here we want $x_1^c(1.5, 2, 10) - x_1^o$, where $x_1^c(1.5, 2, 10)$ is the good 1 quantity that most efficiently achieves old utility 10 at new prices ($p'_1 = 1.5, p_2 = 2$), and $x_1^o = 10$ is old good 1 quantity. Here, since good 1 is still cheaper, we know from (a) that the consumer optimally buys 10 units of it (and no good 2) to most efficiently get old utility 10. So the substitution effect is **zero**. (Intuitively, good 1 is cheaper both before and after the price change, so the consumer doesn't reoptimize at all to maintain utility; no reoptimization means no substitution effect).

- ii. Using your answer to (a), calculate the substitution effect if p_1 increases to 2.5.

Answer: now we want $x_1^c(2.5, 2, 10) - x_1^o$, where again $x_1^o = 10$ is the original good 1 quantity. And x_1^c is the good 1 quantity in the bundle that most efficiently achieves old utility 10 at new prices ($p'_1 = 2.5, p_2 = 2$): here good 1 is more expensive, so by (a), the consumer optimally buys good 2 only, and no good 1. Thus $x_1^c(2.5, 2, 10) = 0$, and so the substitution effect is -10 . (Intuitively, reoptimization leads him to entirely switch from good 1 to good 2, so he quits buying good 1; a decrease of 10).

2. Lagrangians. Suppose you use a Lagrangian to calculate the expenditure function in Q1b, so you set up

$$\mathcal{L} = -p_1 x_1 - p_2 x_2 + \lambda_1 x_1 + \lambda_2 x_2 + \lambda_3 (x_1 + x_2 - u^*)$$

- (a) Explain why this is the Lagrangian, eg what does each part of it represent.

Answer: to form a Lagrangian, first write out the thing you want to maximize: here this is the negative of expenses ($p_1x_1 + p_2x_2$), as minimizing these is the same as maximizing their negative. Then add in every expression that must be nonnegative, each multiplied by its own Lagrange multiplier: we have nonnegativity constraints $x_1 \geq 0$ (so add in λ_1x_1), $x_2 \geq 0$ (so add in λ_2x_2), and the utility constraint is $x_1 + x_2 \geq u^*$ (so add in $x_1 + x_2 - u^*$).

- (b) Write out the optimality conditions, *if you guess that the utility constraint is binding, nonnegativity on good 1 is binding, and nonnegativity on good 2 is non-binding.*

Answer: they are

$$\text{FOC for } x_1 : \frac{\partial \mathcal{L}}{\partial x_1} = 0 \Rightarrow -p_1 + \lambda_1 + \lambda_3 = 0 \quad (1)$$

$$\text{FOC for } x_2 : \frac{\partial \mathcal{L}}{\partial x_2} = 0 \Rightarrow -p_2 + \lambda_2 + \lambda_3 = 0 \quad (2)$$

$$\text{binding } x_1 \geq 0: x_1 = 0 \text{ and } \lambda_1 > 0 \quad (3)$$

$$\text{nonbinding } x_2 \geq 0: x_2 \geq 0 \text{ and } \lambda_2 = 0 \quad (4)$$

$$\text{binding utility constraint} : x_1 + x_2 = u^* \text{ and } \lambda_3 > 0 \quad (5)$$

- (c) Show that you get an invalid solution to (b) if $p_1 < p_2$. And explain why intuitively, the guess was wrong about which nonnegativity constraint binds.

Answer: if we try to solve (1)-(5) in (b), plug $\lambda_2 = 0$ from (4) into (2), to get $\lambda_3 = p_2$. Then plug this into (1) to get $\lambda_1 = p_1 - \lambda_3 = p_1 - p_2$. But if $p_1 < p_2$, this gives a negative value for λ_1 , which is invalid: λ 's must be zero or positive (and here (3) requires λ_1 be in fact positive).

Intuitively, this guess was bad because if $p_1 < p_2$, we know that a consumer with perfect substitute preferences feels that good 1 is a better use of their money. So if they ignored the nonnegativity constraint on good 2, they'd violate it, choosing a negative quantity so as to finance more good 1 (this means we should in fact guess that the constraint $x_2 \geq 0$ is binding). And on the other hand, we know this consumer wants a positive good 1 quantity anyway, so the constraint $x_1 \geq 0$ should be non-binding: this rule won't impact them.

3. Dave has preferences over lotteries which assign probabilities $p = (p_a, p_b, p_c)$ to three possible prizes: an apple, a banana, and a cherry. Suppose that Dave is indifferent between the lottery $p_1 = (1, 0, 0)$ and the lottery $p_2 = (0, \frac{1}{2}, \frac{1}{2})$ and that Dave strictly prefers the lottery $p_3 = (\frac{1}{2}, \frac{1}{2}, 0)$ to the lottery $p_4 = (0, \frac{3}{4}, \frac{1}{4})$. Show that Dave must violate at least one of the von Neumann Morganstern axioms (you don't need to figure out which one).

- easiest answer: we know that if Dave satisfied all axioms, he would be an expected utility maximizer. So if we show these choices are incompatible with EU-maximization, we can conclude that Dave violates at least one axiom. To see that this is indeed the case: for an EU-maximizer to be indifferent between p_1 and p_2 , there must be a utility function defined on prizes apple, banana, cherry such that

$$EU(p_1) = EU(p_2) \Leftrightarrow u(\text{apple}) = \frac{1}{2}u(\text{banana}) + \frac{1}{2}u(\text{cherry})$$

But to prefer p_3 to p_4 , we'd need $EU(p_3) > EU(p_4)$, so

$$\frac{1}{2}u(\text{apple}) + \frac{1}{2}u(\text{banana}) > \frac{3}{4}u(\text{banana}) + \frac{1}{4}u(\text{cherry})$$

which simplifies (subtract $\frac{1}{2}u(\text{banana})$ from both sides and then multiply through by 2) as

$$u(\text{apple}) > \frac{1}{2}u(\text{banana}) + \frac{1}{2}u(\text{cherry})$$

Contradicting the first indifference equation.

- alternate answer: observe that

$$\begin{aligned} p_3 &= \frac{1}{2}(\underbrace{1, 0, 0}_{\equiv p_1}) + \frac{1}{2}(0, 1, 0) \\ p_4 &= \frac{1}{2}\left(\underbrace{0, \frac{1}{2}, \frac{1}{2}}_{\equiv p_2}\right) + \frac{1}{2}(0, 1, 0) \end{aligned}$$

By the independence axiom, p_3 should beat p_4 iff p_1 beats p_2 (we should be able to cancel out the common parts, namely the $\frac{1}{2}$ mixture with $(0, 1, 0)$, and focus on what is different). But Dave has $p_3 \succ p_4$ and $p_1 \sim p_2$, so he violates independence. (Comment: in the lecture slides, I taught independence as $p_1 \succ p_2 \Rightarrow \alpha p_1 + (1 - \alpha)r \succ \alpha p_2 + (1 - \alpha)r$. But one can show that if continuity also holds, it extends to give an indifference version of this axiom, and also the converse; so it is directly stated as “if and only if” in some references).

4. Suppose that with your driving style, you have chance p of getting a speeding ticket which costs you ϕ . Suppose further that the government wishes to hold the expected speeding ticket revenue $p\phi$ equal to a constant, $R \geq 0$. If you are strictly risk-averse, does a reduction in p (subject to the constraint that $p\phi$ be constant) help you or hurt you? (Suggested method: first write out EU, assuming initial wealth w_0 . Differentiate in p , using $\phi = R/p$, to see if EU moves the same or opposite direction to p . Note that for a risk-averse consumer with concave u , we know that for any numbers $a < b$, $u(b) - u(a) < u'(a)(b - a)$).

Answer: expected utility is

$$pu(w_0 - \phi) + (1 - p)u(w_0)$$

Plugging in the constraint $p\phi = R$, we can write this all in terms of p as

$$EU = pu\left(w_0 - \frac{R}{p}\right) + (1 - p)u(w_0)$$

Differentiating in p , we obtain

$$\begin{aligned} EU'(p) &= u\left(w_0 - \frac{R}{p}\right) + pu'\left(w_0 - \frac{R}{p}\right)\left(\frac{R}{p^2}\right) - u(w_0) \\ &= u\left(w_0 - \frac{R}{p}\right) - u(w_0) + \frac{R}{p}u'\left(w_0 - \frac{R}{p}\right) \end{aligned} \quad (*)$$

Applying the hint, with $a = w_0 - \frac{R}{p}$ (the small wealth level) and $b = w_0$ (the large wealth level), concavity buys us

$$\underbrace{u'\left(w_0 - \frac{R}{p}\right)}_{u'(a)} \underbrace{\left(w_0 - \left(w_0 - \frac{R}{p}\right)\right)}_{b-a} > \underbrace{u(w_0) - u\left(w_0 - \frac{R}{p}\right)}_{u(b)-u(a)}$$

This rearranges to yield that expression (*) is positive, i.e. $EU'(p) > 0$. A positive derivative of EU in p means that EU and p move in the same direction, i.e. an increase in p boosts EU , and thus a **reduction in p harms you**.

5. Consider a decision-maker who expects to have a car accident with chance π ; if this occurs, he will incur $\$L$ in damages. He can purchase as much auto insurance, q , as he likes, at a price of p per dollar of coverage: this means that if he pays pq upfront (as the insurance premium), he'll receive a payment of q from the insurance company **if an accident occurs**.

- (a) Write out his expected utility from purchasing insurance level q , assuming a utility-of-wealth function $u(w)$ and initial wealth w_0 .

Answer: with probability π , he has a car accident, for utility $u(w_0 - L - pq + q)$ (from initial wealth w_0 , he loses L in the accident, pays premium pq upfront, then gets coverage q back). With probability $1 - \pi$ there's no accident, so utility is simply $u(w_0 - pq)$ (wealth is initial minus premium). So EU , the probability weighted average of final utilities, is

$$EU = \pi u(w_0 - L - pq + q) + (1 - \pi)u(w_0 - pq)$$

- (b) Show that the optimal level of insurance, q , solves

$$\frac{u'(w_0 - L + (1 - p)q)}{u'(w_0 - pq)} = \frac{p(1 - \pi)}{\pi(1 - p)}$$

Answer: differentiating EU from (a) in the choice variable q , we obtain

$$\pi(1 - p)u'(w_0 - L + (1 - p)q) - p(1 - \pi)u'(w_0 - pq) = 0$$

Which rearranges as the given expression.

- (c) Now assume that $u(w) = 1 - e^{-\alpha w}$, where $\alpha > 0$ (this is known as a “CARA”, or “constant absolute risk aversion”, utility function; α parametrizes risk aversion). Solve explicitly for the optimal insurance, and show that it does not depend on wealth.

Answer: with this utility function, $u'(w) = \alpha e^{-\alpha w}$. Applying this in part (b), the LHS is

$$\frac{\alpha e^{-\alpha(w_0 - L + (1 - p)q)}}{\alpha e^{-\alpha(w_0 - pq)}} = e^{\alpha[w_0 - pq - w_0 + L - (1 - p)q]} = e^{\alpha[L - q]}$$

So the FOC in (b) becomes

$$e^{\alpha(L - q)} = \frac{p(1 - \pi)}{\pi(1 - p)}$$

Taking logs of both sides,

$$\begin{aligned} \alpha(L - q) &= \ln \frac{p(1 - \pi)}{\pi(1 - p)} \\ \Rightarrow L - q &= \frac{1}{\alpha} \ln \frac{p(1 - \pi)}{\pi(1 - p)} \\ \Rightarrow q &= L - \frac{1}{\alpha} \ln \frac{p(1 - \pi)}{\pi(1 - p)} \end{aligned}$$

(this doesn't depend on w_0).